Ayush Carpenter

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GitHub LinkedIn Portfolio LeetCode

Education

Indian Institute of Information Technology, Kota — **Kota, Rajasthan**

B.Tech in Computer Science and Engineering | CGPA: 7.78/10 — *2023 - 2027 (Expected)*

Experience

Gyanama | Frontend Developer Intern — *Remote* | March 2025 - June 2025

- Developed **15+ reusable, responsive UI components** in **React.js, Next.js, and Tailwind CSS**, adopted across **10 product pages** and cutting duplicated UI code by **40%**, establishing a shared component library for the team.
- Collaborated with senior engineers on **frontend architecture**, **state management**, and API data flow, standardizing **3+ new feature modules**.
- Integrated **8+ REST API endpoints** using **React Query** for caching, loading, and error states, reducing redundant network calls by **30%**.

Projects

MediCare | Live Link | GitHub — Next.js 16, React 19, PostgreSQL, WebRTC, Socket.IO, Redis, Docker

- Architected a full-stack telemedicine platform - **JWT auth, RBAC across 3 roles, 39 type-safe Server Actions, 6 Prisma models** - for booking, prescriptions, Razorpay payments and a 7-tool Gemini health assistant; Dockerized.
- Implemented **peer-to-peer video consultations** with **WebRTC** and a custom **Node.js + Socket.IO** signaling server for SDP/ICE negotiation, using **Redis pub/sub** to synchronize sessions across **multiple server instances**.

VersionVista | Live Link | GitHub — React.js, Next.js, MongoDB, simple-git

- Built a **Git repository visualizer using Next.js and React**, parsing and indexing metadata with **simple-git** to replay commit history as an interactive timeline; implemented **file-level and line-level diffs** with side-by-side comparison, handling **1,000+ commits without blocking the UI**.
- Designed a dashboard with a **commit tree, file explorer, and code viewer**, caching repository data in **MongoDB** to reduce analysis time by **approximately 60%**.

CardPlay | Live Link | GitHub — Go, PostgreSQL, WebSockets, Next.js, TypeScript, Docker

- Built a **real-time multiplayer card-game platform** (Monopoly Deal, Trump); verified **exactly-once WebSocket command processing for 8 concurrent actions** and recovery without data loss in restart/reconnect tests.
- Designed **authoritative Go rules engines** behind one pluggable game interface - **89.7% test coverage**, **46 end-to-end tests across 4 live player sessions**.

WealthWise | Live Link | GitHub — React, Next.js, Tailwind CSS, PostgreSQL, Prisma, Gemini API

- Delivered **WealthWise**, an **AI finance platform** (**Next.js 15**, **React 19**, **PostgreSQL**, **Prisma**) with **Clerk auth**, **Recharts** dashboards, **Gemini** receipt OCR, and **Inngest** automated budget alerts and reports.
- Created a **Splitwise-style expense module** with **6 split algorithms**, **group/friend management** with **RBAC**, **pairwise debt netting**, and **settlements** on a **derived-balance ledger** across **10 tables**.

Technical Skills

Languages: C++, JavaScript, TypeScript, Go, Python, Java, SQL

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Redux, Zustand, React Query, Responsive Design

Backend: Node.js, Express.js, REST APIs, WebRTC, Socket.IO, WebSockets, JWT Authentication, Microservices

Databases & ORM: PostgreSQL, MySQL, MongoDB, Redis, Prisma

DevOps & Tools: Docker, Kubernetes, AWS, Git, GitHub, GitHub Actions (CI/CD), Linux, Postman, VS Code

Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

Achievements

Odoo Hackathon: Ranked **Top 7 out of 950+ teams** (**Top 0.7%**) at IIT Gandhinagar.

Competitive Programming: Achieved **LeetCode Knight** with a **1924** rating and finished **Top 500 among 32,000+ participants** (Top **1.6%**) in a Biweekly Contest; **CodeChef 3-Star** and **Codeforces Pupil (1380)**.

Data Structures & Algorithms: Solved **1,050+** problems on LeetCode (**850+**) and GeeksforGeeks (**200+**), spanning graphs, dynamic programming, trees, binary search, and greedy algorithms.